Sports vs Politics Classifier

Pragna Sree Muvva
Roll Number: B23CM1025

1 Introduction

The text classification is a fundamental task in Natural Language Understanding(NLU), which involves automatically classifying text documents into predetermined categories. For this Problem, we implement a binary text classification model that classifies news documents into two categories: **Sports** and **Politics**.

This task has real-world applications, especially within news aggregation sites, content filtering tools, recommendation tools, and even automated journalism tools. The aim of this problem is to explore different feature sets and to implement three machine learning algorithms to find out which approach is best suited to solve this text classification task.

2 Dataset Collection and Description

The dataset used in this project is AG News Classification dataset, a structured dataset for news articles stored in `df_file.csv`. The dataset was originally in five different classes, such as sports, business, politics, sci-fi/tech, etc. But in this project, we only require two classes:

- Label 0: POLITICS
- Label 1: SPORTS

After filtering, the dataset contains approximately:

- 417 Politics articles
- 511 Sports articles

This ensures a reasonably balanced binary classification dataset. Each sample contains: A full news article (text) and A category label. The dataset includes real news content, making the classification problem realistic and non-trivial.

3 Preprocessing

Before training the models, basic preprocessing was applied to normalize the text data. No aggressive stopword removal or stemming was performed to ensure meaningful contextual words are maintained. The following preprocessing was applied to the text data before training the models:

- Conversion to lowercase
- Removal of punctuation and numerical characters
- Removal of special symbols

The cleaning function used:

```
text = text.lower()
text = re.sub(r"[^a-z\s]", " ", text)
```

4 Feature Engineering

Feature engineering is an important aspect of text classification problems because machine learning models cannot directly process text. Text needs to be converted into a numerical representation. For this project, three different methods of feature extraction were used: Bag of Words, TF-IDF, and TF-IDF with Bigrams.

4.1 Bag of Words (BoW)

The Bag of Words model is one of the easiest and most commonly used methods to represent the text data. In the Bag of Words model, the documents are converted into vectors with the frequency of the words in the document. The total number of words in the vocabulary is obtained by taking the union of all the words from the training documents.

If the total number of unique words in the vocabulary is V, the document d is represented as a vector X_d with V features:

$$X_d = (x_1, x_2, ..., x_V)$$

where x_i denotes the frequency of the i-th word in the document.

Advantages of Bag of Words:

- It is the easiest and fastest method.
- It is the easiest method to implement.
- Surprisingly good results are obtained using the Bag of Words model.

Limitations:

- The context and the order of the words are not taken into account.
- The Bag of Words model assumes that all the words are equally important.
- The Bag of Words model is dominated by the frequently occurring words such as "the," "is," etc.

4.2 TF-IDF (Term Frequency - Inverse Document Frequency)

TF-IDF is an improvement over the Bag of Words model in that words are weighted according to their importance in the entire corpus. The intuition behind the TF-IDF model is that words that appear more frequently in a given document but less frequently in the entire corpus are more informative for classification. The TF-IDF score for a term t in document d is computed as:

$$TFIDF(t,d) = TF(t,d) \times \log\left(\frac{N}{DF(t)}\right)$$

Where:

- $TF(t,d)$ = term frequency of word t in document d
- N = total number of documents
- $DF(t)$ = number of documents containing term t

The term frequency component measures the word's frequency in a document, while the inverse document frequency component controls for words that are too common in the set of documents.

Advantages of TF-IDF:

- It reduces the effect of common but uninformative words.
- It emphasizes discriminative words.
- It often performs better than the BoW approach in terms of classification accuracy.

In the context of this project, TF-IDF helps distinguish domain-specific terms such as "election", "parliament", "goal", and "tournament".

4.3 TF-IDF with Bigrams

While unigram models (single words) capture individual terms, they fail to capture local context. For example: "prime minister", "world cup", "transfer market"... These phrases carry stronger semantic meaning than individual words alone. To capture such contextual patterns, we extended TF-IDF to include bigrams using:

```
TfidfVectorizer(ngram_range=(1,2))
```

This configuration enables the model to look at unigrams and bigrams at the same time.

Benefits of using bigrams:

- It can capture short phrase-level semantics.
- It can improve discrimination between classes.
- It can represent political entities and sports events better.

However, the addition of bigrams also adds dimensions to the feature space, which could increase computational costs. But, fortunately, linear classifiers like SVM can cope with high-dimensional sparse data.

4.4 Comparative Perspective

The three feature representations differ in complexity and expressive power:

- **BoW**: Simple frequency-based representation.
- **TF-IDF**: Weighted representation emphasizing informative words.
- **TF-IDF + Bigrams**: Context-aware representation capturing short phrases.

Evaluating all three enables us to analyze how feature engineering impacts classification performance in the Sports vs Politics task.

5 Machine Learning Models

In order to test the efficiency of various learning strategies for text classification, three of the most widely used machine learning algorithms were implemented and compared: Multinomial Naive Bayes, Logistic Regression, and Support Vector Machine (SVM). These models represent different theoretical approaches to the problem.

5.1 Naive Bayes

Multinomial Naive Bayes is a probabilistic classifier based on Bayes' Theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where:

- C represents the class (SPORTS or POLITICS),
- X represents the feature vector (word frequencies or TF-IDF values).

This means that the model assumes conditional independence of the features, i.e., the occurrence of one word is independent of the occurrence of another word given the class label. This assumption is not very realistic in the case of natural languages, but the Naive Bayes classifier does surprisingly well in text classification problems.

This model is especially suited for document classification because the multinomial model counts the occurrence of words. Its advantages are:

- Fast training and prediction
- Low computational complexity
- Strong baseline performance

However, its main limitation is the independence assumption, which may reduce performance when features are highly correlated.

5.2 Logistic Regression

Logistic Regression is a discriminative linear classifier that models the probability of a class using the logistic (sigmoid) function:

$$P(y = 1|X) = \frac{1}{1 + e^{-(w^T X + b)}}$$

Unlike Naive Bayes, Logistic Regression does not assume independence between features. Instead, it directly learns the weights of each feature that maximizes the probability of correct classification.

In high-dimensional text classification, Logistic Regression can perform well because:

- It can handle correlated features better than Naive Bayes.
- It can provide calibrated probabilities.
- Regularization can also be used to prevent overfitting.

As a result, Logistic Regression can perform better than Naive Bayes in terms of accuracy in many real-world text classification tasks.

5.3 Support Vector Machine (SVM)

"Support Vector Machine" is a margin-based classifier that tries to find the optimal separating hyperplane between two classes. The goal of a linear Support Vector Machine is to maximize the margin between the nearest data points (support vectors) of the two classes.

Mathematically, the optimization problem solved by the Support Vector Machine is given by:

$$\min_{w,b} \frac{1}{2}||w||^2$$

subject to appropriate classification constraints.

Support Vector Machine is a good classifier when it comes to text classification because:

- The dimensionality of the text data is usually high.
- The data is sparse.
- Linear decision boundaries are good in high-dimensional space.
- The Support Vector Machine maximizes the margin between classes.

In fact, the Support Vector Machine usually outperforms other linear classifiers when it comes to document classification tasks.

5.4 Model Comparison Perspective

The three models represent different paradigms:

- **Naive Bayes** - Generative model
- **Logistic Regression** - Discriminative probabilistic model
- **SVM** - Margin-based classifier

Comparing these models allows us to analyze how different assumptions and optimization strategies affect classification performance in the Sports vs Politics task.

6 Experimental Setup

The dataset was split into:

- 80% Training data
- 20% Testing data

A fixed random seed was used for reproducibility.

Each feature representation (BoW, TF-IDF, TF-IDF + Bigrams) was evaluated with all three classifiers.

7 Results and Quantitative Comparison

Feature Method	Naive Bayes	Logistic Regression	SVM
Bag of Words	1.00	0.97	0.98
TF-IDF	1.00	0.98	1.00
TF-IDF + Bigrams	1.00	0.99	1.00

Table 1: Accuracy Comparison of Models

From the results, we observe:

- While achieving perfect accuracy might indicate strong class separability, it also suggests that the dataset may contain highly domain-specific vocabulary. In real-world scenarios with more nuanced or overlapping content, performance may be lower.
- TF-IDF performs better than Bag of Words.
- Adding bigrams further improves performance.
- SVM achieved near-perfect performance.

8 Confusion Matrix

For the best performing model (SVM with TF-IDF + Bigrams), the confusion matrix showed strong separation between the two classes.

$$\begin{bmatrix} \text{True Politics} & \text{False Sports} \\ \text{False Politics} & \text{True Sports} \end{bmatrix}$$

Most misclassifications occurred in borderline cases where sports articles referenced political entities or vice versa.

9 Limitations

Despite strong performance, the system has several limitations:

- It only works with binary classification.
- It cannot grasp deep semantics or sarcasm.
- It may not work well with unseen domains.
- It doesn't have advanced linguistic preprocessing such as stemming or lemmatization.
- It is sensitive to domain shift.

10 Conclusion

This classifier has successfully implemented and compared three machine learning models for classification of news articles into Sports and Politics classes.

Key findings:

- TF-IDF outperforms simple Bag of Words.
- Including bigrams improves contextual understanding.
- SVM achieved the best overall performance.

The results demonstrate that classical machine learning techniques remain highly effective for structured text classification tasks.

There are several possibilities for future work, including:

- Deep learning models (e.g., LSTMs, Transformers)
- Larger datasets
- Multi-class classification